

CHAEWAN CHUN

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RESEARCH EXPERIENCE & SELECTED PROJECTS

Ph.D. Research Scientist Intern

Jun – Aug 2026

Air Force Research Laboratory (AFRL)

- Conducted research and development; technical details are currently restricted from public disclosure.

Conversational Audio Fact-Checking & Spoken-Media Integrity

Jan 2025 – Present

Pennsylvania State University

- Developed MAD/MAD2, the first conversational audio fact-checking benchmarks (1,000 dialogues; 3,368 check-worthy claims; ~10h aligned audio), enabling rigorous evaluation of multi-turn spoken claims beyond isolated text settings.
- Engineered an end-to-end pipeline transforming seed claims into claim-aligned spoken dialogues via LLM-based dialogue synthesis, WhisperX timestamp grounding, and MoonCast speech synthesis.
- Designed context-controlled evaluation regimes, facilitating a realistic assessment of how conversational context and modality choices impact verification accuracy.

Amazon Nova AI Challenge 2026

Feb 2026 – Present

Amazon & Pennsylvania State University

- Selected as a member of a global top-10 university team for the 2026 Amazon Nova AI Challenge — Trusted Software Agents track, a competitive international AI competition focused on building and evaluating secure, agentic AI systems using Nova Forge.

Computational Brushstroke Analysis & Feature Engineering

Aug 2022 – Dec 2024

Pennsylvania State University

- Developed a framework to quantify artistic style by extracting brushstroke *flow* via structure-tensor fields and streamline tracing, transforming static image patches into computational representations of dynamic movement.
- Engineered high-dimensional distributional feature sets (e.g., curvature smoothness, directional variability, and segment length) to characterize the gestural expressiveness of Claude Monet's paintings.
- Integrated these geometric features into a novel C2A (Composition to Attribute) deep neural network, enabling the first systematic classification of “abstract” vs. “non-abstract” brushwork with strong performance (mean accuracy ≈ 0.91 under LOPO-CV).

3D Motion Dynamics & Behavioral Analysis Framework

May 2023 – Dec 2024

Pennsylvania State University

- Developed a 3D pose-based time-series clustering framework to quantify motion dynamics from video clips, investigating the correlation between physical movement patterns and emotional states.
- Engineered an Active Learning pipeline featuring model-based uncertainty sampling to prioritize rare emotion classes (e.g., disgust) for expert annotation, significantly reducing labeling costs while enhancing model robustness against class imbalance.

Volvo Trucks – Fuel-Efficiency Prediction & Configuration Insights

Aug – Dec 2021

Pennsylvania State University

- Prepared and modeled a high-dimensional fleet dataset (181,805 vehicles $\times$ 145 features) to predict fuel efficiency from truck configuration and operational attributes.
- Translated model feature effects into three configuration recommendations for improving customer fuel efficiency, delivered to Volvo stakeholders.

EDUCATION

The Pennsylvania State University *University Park, PA* Aug 2022 – Present
Ph.D. candidate in **Informatics** | Advisor: [Dr. Dongwon Lee](#) GPA: 3.94/4.0

The Pennsylvania State University, *University Park, PA* Aug 2018 – May 2022
Schreyer Honors College, Dean's List (every semester) GPA: 3.91/4.0
B.S. in **Computer Science**, *Magna Cum Laude* | B.S. in **Mathematics**, *Cum Laude* | Minor: **Statistics**
HONORS THESIS: [Robust Image Classification based on Pixel Importance](#) (Advisor: [Dr. Jia Li](#)).

PUBLICATIONS

Chaewan Chun, Delvin Ce Zhang, and Dongwon Lee. “Context-Aware Multimodal Claim Verification in Spoken Dialogues.” *Under review at the 2nd Speech and Audio Language Models Workshop (SALMA), EMNLP*, 2026.

Meruyert Aristombayeva, Jason Lucas, **Chaewan Chun**, and Dongwon Lee. “Lost in Speech: Trilingual Spoken Hallucination Detection Across Audio and Transcripts.” *Under review at the 2nd Speech and Audio Language Models Workshop (SALMA), EMNLP*, 2026.

Chaewan Chun, Meruyert Aristombayeva, Jiyoung Choi, Mahjabin Nahar, Delvin Ce Zhang, and Dongwon Lee. “TRILOGUE: A Trilingual Spoken Dialogue Fact-Checking Benchmark with Evidence and Paired Audio.” *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

Jiaxi Yang, **Chaewan Chun**, Jason Lucas, Yuchen Yang, and Dongwon Lee. “AOR-Bench: Do Large Audio Language Models Over-Refuse Pseudo-Harmful Queries?” *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

Chaewan Chun, Delvin Ce Zhang, and Dongwon Lee. “When Misinformation Speaks and Converses: Rethinking Fact-Checking in Audio Platforms.” *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. **Oral presentation.**

Lizhen Zhu, **Chaewan Chun**, Kathryn Brown, and James Z. Wang. “Mapping the Flow of Painterly Gesture.” *Patterns*, vol. 7, no. 3, article 101516, 2026.

Jia Li, **Chaewan Chun**, Kathryn Brown, and James Z. Wang. “Computational Investigation of Abstraction in Claude Monet’s Water Lilies Through Brushstroke Analysis.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 6, pp. 7129–7146, 2026.

Chaewan Chun, Lysandre Terrisse, Delvin Ce Zhang, and Dongwon Lee. “MAD: A Benchmark for Multi-Turn Audio Dialogue Fact-Checking.” *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS)*, accepted as a working paper, 2025.

AWARDS & GRANTS

Selected Future Leader of AI, <i>Doctoral Consortium, ACM AI Leadership Summit</i>	Aug 2026
Fund for Excellence in Graduate Recruitment Award, <i>academic achievement, Penn State</i>	Jun 2022
Paul MacDonald Open Doors Scholarship, <i>leadership and academic achievement, Penn State</i>	Nov 2021
Leadership Scholarship, <i>leadership engagement in Women in Engineering, Penn State</i>	May 2021
Student Engagement Network Innovation Grant, <i>summer research, Penn State</i>	Apr 2021
Evan Pugh Scholar Award, <i>academic achievement, Penn State</i>	Apr 2020

TEACHING, LEADERSHIP, AND SERVICE

LEADERSHIP:	Teaching Assistant, Interdisciplinary Research Design for IST (IST 501) Learning Assistant, Programming Language Concepts (CMPSC 461) Administrative Assistant - data analytics Engineering Orientation Network Leader/Mentor Penn State Peer Mentor Collective Facilitated Study Group Coordinator, Women in Engineering.
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